THE JOURNAL OF PHYSICAL CHEMISTRY C

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Attractive Interactions between DNA−Carbon Nanotube Hybrids in Monovalent Salts

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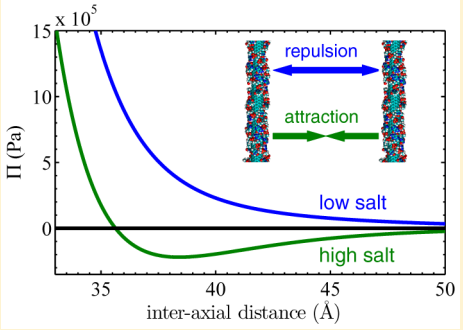
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Supporting Information

ABSTRACT: DNA−carbon nanotube (DNA-CNT) hybrids are nanometer-sized, highly charged, rodlike molecules with complex surface chemistry, and their behaviors in aqueous solutions are governed by multifactorial interactions with both solvent and cosolutes. We have previously measured the force between DNA-CNTs as a function of their interaxial distance in low monovalent salts where interhybrid electrostatic repulsion dominates. The characteristics of DNA-CNT forces were further shown to closely resemble that of double-stranded DNA (dsDNA) in low salts. However, contrasting behaviors emerge at elevated monovalent salts: DNA-CNT condenses spontaneously, whereas dsDNA remains soluble. Here we report force−distance dependencies of DNA-CNTs across wide-ranging monovalent salt concentrations. DNA-CNT force curves are observed to deviate from dsDNA curves above 300 mmol/L NaCl, and the deviation grows with increasing salts. Most notably, DNA-CNT forces become net attractive above 1 mol/L NaCl, whereas dsDNA forces are repulsive at all salt concentrations. We further discuss possible physical origins for the observed DNA-CNT attraction in monovalent salts, in consideration of the complex surface chemistry and unique polyelectrolyte properties of DNA-CNT hybrids.



INTRODUCTION

DNA and carbon nanotubes represent two of the most diverse members of their respective biological and inorganic worlds. Single-stranded DNA are flexible chains of a set of four nucleotides, and their combinatorial sequences, in astronomical numbers, encode the blueprint of life and afford practically unlimited variations as the favorite building blocks for nanobiotechnology.[1,2] Single-wall carbon nanotubes (CNTs) are rigid hollow cylinders enclosed by one-atom-thick graphene sheets, and the myriad rolling vectors of graphene give rise to chirality-dependent CNT structures.[3] As such, noncovalent conjugates of DNA and CNTs are promising bioabiotic hybrids of extraordinary biological, chemical, and physical properties.

Early studies of DNA-CNT hybrids focused on overcoming two major barriers of the CNT technology: water-insolubility and inhomogeneous chirality distribution of as-synthesized CNTs. Aqueous dispersion of CNTs is a significant step because it allows uses of effective solution-based processing techniques such as characterization and assembly.[4] Likewise, sorting by CNT chirality is essential for exploring the chirality-specific electronic, optical, and thermal properties of CNTs.[5] Remarkably, DNA-assisted dispersion and sorting of CNTs have attained CNT solutions among the highest concentrations and, most importantly, a growing list of CNT types of highest single-chirality purities.[6−8]

Given the chemically homogeneous and hydrophobic CNT surface, its high affinity to DNA necessarily arises from surface stacking with hydrophobic DNA motifs (i.e., aromatic planes of nucleobases) via a combination of interdependent hydrophobic, electronic $\pi-\pi$, and van der Waals interactions.[9,10] The highly charged DNA backbones exposed outside subsequently confer high solubility. Initial sorting studies utilized ion-exchange chromatography (IEX) to separate DNA-CNT hybrids by CNT chirality.[6−8] It was observed that CNTs of specific chirality are best sorted with DNA of specific sequence, indicating peculiar couplings between CNT chirality and DNA sequence in determining the hybrid surface characteristics (e.g., charge density and hydration strength). Knowledge of the exact DNA structure on the CNT surface is key to unraveling the underlying mechanisms but remains unknown. Recently, a breakthrough in CNT sorting demonstrated that DNA-CNT hybrids partition differentially in neutral-polymer-based aqueous two-phase (ATP) systems,[11] and such partition is strongly modulated by DNA sequence, CNT chirality, ATP composition, and salt conditions. It is worth noting that the ATP-based CNT sorting is also applicable to surfactant-dispersed CNTs,

Received: May 7, 2016
Revised: June 1, 2016
Published: June 2, 2016

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DOI: 10.1021/acs.jpcc.6b04623
J. Phys. Chem. C 2016, 120, 13831−13835

though the exact partition parameters and outcomes differ.[12−15] While it is apparent that electrostatic interactions play an important role in IEX separation, the mechanism of ATP separation highlights the importance of DNA-CNT solvation energy which is under delicate control by a number of internal (DNA and CNT) and external (e.g., polymer, salt, and pH) factors.[11] Given the ability to readily modulate these factors and quantify their effects through CNT partitioning, the system of DNA-CNT hybrids presents a unique opportunity to probe the intricate interactions between an inorganic surface, a charged biopolymer, and their aqueous and ionic environment.

We have previously studied the electrostatic and hydration interactions of DNA-CNT hybrids in low monovalent salts (up to 500 mmol/L NaCl) by measuring the interhybrid force as a function of the interhybrid distance.[16,17] The main finding is that DNA-CNT hybrids exhibit force−distance dependencies very similar to that from double-stranded DNA (dsDNA) in low salts, once their difference in diameter is taken into account. Consistent with their highly charged nature, electrostatic repulsion was observed to dominate at low salts. At close surface−surface separations (<1 nm), the force required to dehydrate hybrid surfaces becomes significant and gives rise to the so-called hydration forces as previously observed in dsDNA interactions.[18−20] However, despite the resemblances in low salts, disparate behaviors are observed in high salts: DNA-CNT hybrids spontaneously condense at >1 mol/L monovalent salts,[21,22] while dsDNA remains soluble (e.g., up to 3 mol/L NaCl). This qualitative difference indicates a source of attractive force present in DNA-CNT only, and a new mode of DNA-CNT interaction is suggested. To this end, this study aims to quantify the force characteristics in high monovalent salts and to investigate likely physical origins for DNA-CNT attraction. By measuring the force curves of DNA-CNT and dsDNA under identical conditions, we report conclusive evidence for the presence of attraction in DNA-CNT in monovalent salts. We further parametrize the measured force−distance dependencies based on the hydration force formalism and estimated the magnitudes of the DNA-CNT attraction. Possible mechanisms for DNA-CNT attraction are discussed in light of the distinct surface chemistry and polyelectrolyte properties of DNA-CNT.

MATERIALS AND METHODS

Preparation of DNA-CNT hybrids followed the protocols as described in refs 16 and 23, and force measurements were carried out with the osmotic stress method (OSM).[16,24] We provide key specifics below, and additional details can be found in the Supporting Information. Single-wall CNT (CoMoCAT SG65i) was obtained from South West Nano Technologies, and single-stranded $(GT)_{20}$ DNA was obtained from Integrated DNA Technologies; genomic dsDNA (Salmon, >10 kilo base pairs) was purchased from Sigma. To probe intermolecular forces at close separations, condensed pellets of DNA-CNT or dsDNA were first obtained via polymer-induced formation of liquid crystalline phases in low salts. It is worth noting that low salts (<500 mmol/L) were necessary for DNA-CNT to form liquid crystalline phases.[17] Each pellet was then equilibrated against an osmolyte solution exerting an external osmotic pressure over the liquid crystalline phase. Because of pressure balance, the force/pressure between DNA-CNTs or dsDNAs was given by the osmotic pressure, and the interaxial distance was subsequently measured with X-ray diffraction. The force−distance curve was obtained by varying the osmotic pressure via osmolyte concentration.

RESULTS

The highly charged nature of DNA-CNT hybrids results in strong electrostatic interactions that dominate in low salt concentrations. Increasing salt concentration consequently provides a convenient way to weaken electrostatic forces and makes it possible to discern the existence of other types of interactions. We show the force between DNA-CNT hybrids as a function of interaxial distance in Figure 1 under salt conditions ranging from 300 mmol/L to 2 mol/L NaCl, noting that the force is shown as the osmotic pressure that is directly varied in experiments. The force−distance curves of dsDNA under the same salt conditions are also shown to illustrate the force characteristics from a charged and fully hydrated surface. The force curves in even lower salts of 50 and 150 mmol/L NaCl have been analyzed previously,[16] where DNA-CNT and dsDNA show the same shapes once differences in their diameters are taken into account. Given that a different batch of CNT (CoMoCAT SG65) was used in our earlier study,[16] the results are reproduced in Figure S2a-d for the CNT (CoMoCAT SG65i) studied herein, noting that a difference of 2 Å in average DNA-CNT diameters was found. Upon increasing salts, both sets of curves show decreasing forces, consistent with salt screening of the electrostatic repulsion. However, the shapes of the DNA-CNT and dsDNA curves begin to differ. Qualitatively, all the dsDNA force curves can be described by a fast-decaying repulsion dominating at the short-range and a slow-decaying repulsion at the long-range, giving rise to "concave" shapes under all salt conditions. In contrast, the DNA-CNT force curves above 1.0 mol/L NaCl are "convex": while the fast-decaying short-range repulsion persists, a long-range attraction force necessarily exists to explain the "downturn" of the curves with increasing interaxial distances.

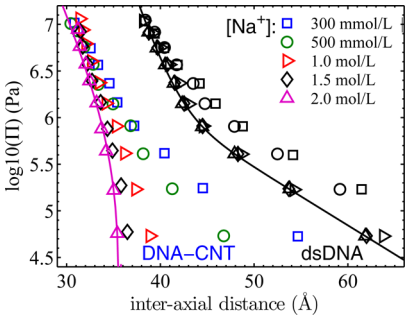


Figure 1. Force−distance curves of the DNA-CNT hybrid and dsDNA at different salts. The same symbol is used for a specific Na^+ concentration as denoted in the legend. The dsDNA curves in black are shown with a positive offset of 13 Å in interaxial distance to aid visualization. Solid lines are guides to the eye to illustrate the "concave" and "convex" shapes. Note that the 300 and 500 mmol/L NaCl DNA-CNT curves have been shown previously,[17] though no comparison or analysis with dsDNA curves was carried out.

We next seek to provide an analytical formulation of the measured DNA-CNT forces and further extract the magnitude of the attraction between DNA-CNTs. In our previous analysis of DNA-CNT forces in low salts (50 and 150 mmol/L NaCl),[16] we showed that the hydration force formalism,[24] proposed by Parsegian and Rau for dsDNA forces at the last nanometer of separation,[25] is able to quantitatively describe the

DOI: 10.1021/acs.jpcc.6b04623
J. Phys. Chem. C 2016, 120, 13831−13835

DNA-CNT forces in low salts. The notion of hydration force arises from the observations that measured inter-dsDNA forces at close separations are far greater than expected charge-driven double-layer interactions and that the force characteristics suggest dominant contributions from the hydration layers. Physically, the hydration force between two approaching surfaces can be separated into the desolvation of each surfaces and the interaction between their respective hydration layers.[25] The energetic cost of desolvation gives rise to an "image-charge-like" repulsive term ($f_d e^{-d/(\lambda/2)}$), with a decay length of $\lambda/2$, where λ is the water correlation length ~4−6 Å; and the interaction term ($f_i e^{-d/\lambda}$) has a decay length of λ and can be attractive or repulsive depending on whether the interaction strengthens or weakens respective hydration structures. In the case of charged surfaces, an additional double-layer interaction term ($f_e e^{-d/\lambda_D}$; λ_D is Debye length) is included. Note that Figure 1 shows that the dsDNA forces are essentially independent of salt concentrations in high salts ≥1 mol/L, evidencing the importance of hydration contributions. Because of the similarity between λ (~4−6 Å) and λ_D in intermediate salt conditions (e.g., 500 mmol/L NaCl gives λ_D ~ 4.3 Å), an effective interaction, $f_{eff} e^{-d/\lambda_{eff}}$, is used empirically to characterize the long-range force. As a result, intermolecular forces are described by two exponentials as $F(d) = f_d e^{-d/(\lambda/2)} + f_{eff} e^{-d/\lambda_{eff}}$ with f_d, f_{eff}, and λ_{eff} as the unknowns determined by fitting (see the Supporting Information for more detailed discussions on the hydration force formalism).

The DNA-CNT forces are then analyzed based on the double exponential force. Qualitatively, the convergence of the force−distance curves at close separations (Figure 1) corroborates the presence of a common hydration repulsion dominating the short-range forces independent of salts. Forces at longer distances show clear salt dependence indicating significant electrostatic double-layer interactions. Quantitatively, the fits using double exponential forces are shown in Figure 2. Note that, while the dsDNA of genomic origin is polydisperse in length, the forces measured here are not subject to the length effect because the inter-dsDNA distances of interest here are much shorter than the minimum dsDNA length (<4 nm versus 10 μm). In the fits, water correlation length, λ, is fixed at 4 Å and λ_{eff}, f_d, and f_{eff} are allowed to vary. Of the two exponential forces, the hydration repulsion of $\lambda/2$ decay length dominates the short-range forces. It is shown to describe very well the short-range behaviors of DNA-CNT forces up to 500 mmol/L NaCl (Figure 2a) and all dsDNA forces (Suppl. Figure S4). However, from 1 mol/L NaCl (Figure 2b), DNA-CNT experimental curves show an "upturn" relative to the fits upon decreasing separation, and the upturn increases with salt concentration. It should be noted that salt-induced structural changes of DNA-CNT hybrids have been reported only for divalent cations.[26] The DNA-CNT hybrids studied here are known to be stable in monovalent salts, e.g., ion exchange chromatography used a 1 M NaSCN for separation.[8] While we do not know the origin of this salt-induced additional repulsion, this observation suggests that the hydration force formalism no longer suffices to explain the short-range force behaviors in high salts and that this additional repulsion is of very short-range (<2 Å). On the other hand, the second exponential force of longer decay length ($f_{eff} e^{-d/\lambda_{eff}}$) appears to fit well the forces at longer ranges. Overall, with the exception of an unknown salt-induced repulsion in high salts (>500 mmol/L NaCl), the double exponential forces are shown

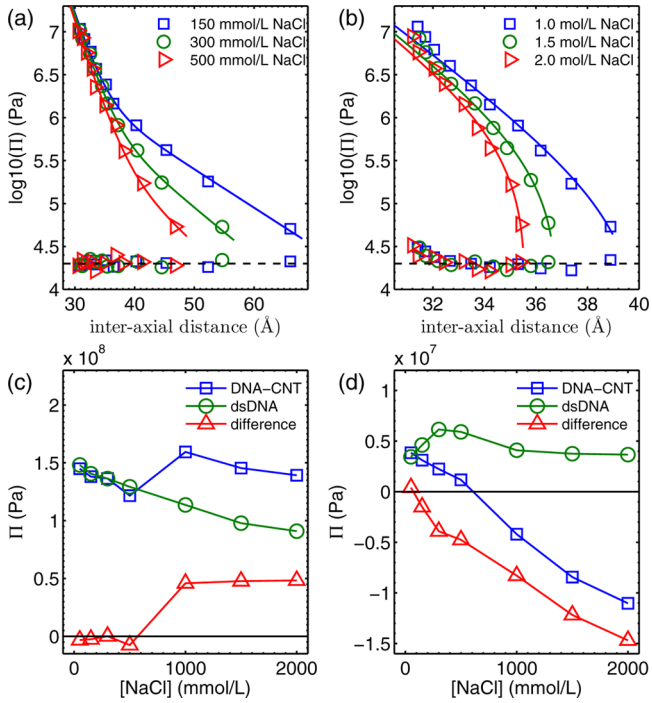


Figure 2. Fits to the force−distance curves of DNA-CNT hybrids using the double-exponential forces as described in the text. (a) Low salt data (150, 300, and 500 mmol/L NaCl). Symbols are experimental data as denoted in the legend; the corresponding fits are shown in the same color as solid lines, and the differences between experimental data and fits are shown below with an offset indicated by the dashed line. (b) High salt data (1000, 1500, and 2000 mmol/L NaCl). Annotations are as described in panel a. Note that the differences shown below indicate the "upturn" in the experimental data. (c) The magnitude of the first exponential force at the molecular surface, i.e., the hydration repulsion ($f_d e^{-d/(\lambda/2)}$) where d is the diameter. The plotted magnitudes correspond to the forces at the interaxial distance of 25.5 Å for the DNA-CNT hybrid and 20 Å for dsDNA. The 5.5 Å difference in diameter is determined based on the matches at low salts (see Figures S2 and S3 for the matches). (d) The magnitude of the second exponential force ($f_{eff} e^{-d/\lambda_{eff}}$) at the molecular surface for both DNA-CNT hybrids and dsDNA.

to be able to quantitatively describe DNA-CNT and dsDNA forces.

With both DNA-CNT and dsDNA forces parametrized in the form of double exponentials, we compare their force parameters in Figure 2c,d. The short-range hydration repulsion ($f_d e^{-d/(\lambda/2)}$) corresponds to the energetics of dehydration and is directly linked with the solvation energy of the hybrid. Figure 2c shows its magnitude at the molecular surface to account for the different diameters of DNA-CNT (25.5 Å) and dsDNA (20 Å). Notably, the force magnitudes are very similar between DNA-CNT and dsDNA at ≤500 mmol/L NaCl, indicating similar densities of surface charged and polar groups (i.e., similar levels of surface hydration). At salts ≥1 M NaCl, DNA-CNT shows force magnitudes larger than those of dsDNA likely because of the additional salt-induced short-range repulsion discussed earlier. As for the second exponential force ($f_{eff} e^{-d/\lambda_{eff}}$), the decay length is observed to decrease monotonically with salt as expected from the decreasing Debye length of the electrostatic contribution (Figure S5). The force magnitude from dsDNA shows no clear trend with salt concentration, suggesting that the effect of salt screening is largely captured by the decrease of decay length. However, the

DOI: 10.1021/acs.jpcc.6b04623
J. Phys. Chem. C 2016, 120, 13831−13835

force magnitude from DNA-CNT decreases monotonically with salt (Figure 2d). Moreover, from 1 mol/L NaCl, the second exponential force of DNA-CNT turns from repulsive to attractive (i.e., with negative magnitudes), while dsDNA shows repulsive second exponential forces in all salt concentrations.

DISCUSSION AND CONCLUSION

Through measurements and analyses of the force−distance dependencies of DNA-CNT hybrids over a broad range of salt concentrations, we provide quantitative parametrizations and physical descriptions of the interhybrid forces. We consider the salt-induced transition from repulsive to attractive interhybrid long-range interaction as the most significant experimental observation of this study. Parameterization using double exponentials not only provides a convenient form of interaction potentials but also sheds lights on the physical nature of the interactions by separating the forces operating at different distance ranges and arising from different physical origins. In direct comparison with dsDNA forces, DNA-CNT forces exhibit two distinct regimes dependent on salt concentration. In low salts (<500 mmol/L NaCl), DNA-CNT and dsDNA show the same force characteristics: dominant hydration repulsion at the shortest separations and double-layer electrostatic repulsion at long distances. Both force behaviors can be attributed to their large charge densities. However, qualitatively different forces are observed between DNA-CNT and dsDNA in high salts (>1 mol/L NaCl). While dsDNA forces show only slight decreases in the double-layer electrostatic repulsion, DNA-CNT forces show qualitative changes: an excess salt-induced repulsion at the short-range and a transition from repulsion to attraction at the long-range, with the latter leading to the macroscopic salt-induced DNA-CNT condensation previously observed.[17] Such disparate force behaviors between DNA-CNT and dsDNA in high salts suggest the existence of inter-(DNA-CNT) interactions of a physical nature different from the hydration and double-layer interactions in low salts. While the concurrence of short-range repulsion and long-range attraction suggests the same physical origin, there is no conclusive evidence because of the limited salt concentrations studied.

In consideration of the chemical makeup and physical attribute of DNA-CNT, we next discuss several possible physical origins for its force behaviors in high salts, with a particular emphasis on the long-range attraction leading to macroscopic condensation. One possibility is that the long-range attraction is directly mediated by monovalent salts, i.e., of electrostatic origin. Given that DNA-CNT has the same charged and polar groups as dsDNA which exhibits no attraction, a new mode of hybrid-ion interaction is called for. One distinctive feature of DNA-CNT as a polyelectrolyte is its much larger persistence length (>32 μm[27]) in comparison with dsDNA (~50 nm). It is conceivable that this rigidity facilitates ordered arrangements of condensed ions around DNA-CNT surfaces, which consequently enhances spatial correlations between opposing hybrid surfaces and results in attraction. These ordered ions, presumably in high salts only, also increase the "apparent" hybrid diameter giving rise to a concurrent short-range repulsion. This ideal rigidity further suggests that DNA-CNT may make an excellent model system for studying electrostatics of rodlike polyelectrolytes, perhaps even better than dsDNA. It is worth noting that the phenomenon of overcharging, demonstrated for both single- and double-strand DNA in multivalent salts,[28−30] is not expected to apply here because monovalent salts are used in this study. The other possibility is that the long-range attraction is of nonelectrostatic nature. As such, the attraction exists in low salts as well but is overwhelmed by the strong electrostatic repulsion therein. One likely source is van der Waals (vdW) interaction between DNA-CNTs. The strength of vdW forces is determined by the dielectric response discontinuities between CNT and solvent and is characterized by the Hamaker coefficient, H. French et al.[31] showed that the magnitude of vdW forces between two parallel CNTs per angstrom length is given as $\frac{\sqrt{a}}{16t^{5/2}}H$, where a is the CNT radius and t is the surface separation. Taking a = 3.74 Å, H = 70 zJ for (6,5) CNT in water medium,[31] and t = 30 Å (i.e., an interaxial distance of 37.5 Å around the middle of measured ranges), this gives a force of 0.017 pN and pressure of 4.5×10^4 Pa. Compared with dsDNA forces at the same surface−surface separation, the CNT vdW force is 20−100 times smaller under all salt conditions. It is thus unlikely that CNT vdW interaction is the cause for the attraction between DNA-CNT hybrids. Another possible source of attraction is hydrophobic interaction between exposed nonpolar groups of DNA-CNT hybrids. Such nonpolar groups may include uncovered CNT surfaces and DNA nucleobase basal planes that are exposed to solvent, as revealed in a number of computer simulation studies.[10,32] Furthermore, DNA absorption on CNT surfaces may change with salt concentration,[33] which would lead to salt dependence of interhybrid interactions. The exact chemistry and surface pattern of these nonpolar groups necessarily depend on the DNA structure on the CNT surface which, however, is unknown experimentally.

The results from this study have potential to guide practical applications particularly involving molecular self-assemblies of DNA-CNTs, where the analytical parametrization of interhybrid forces further allows the development of predictive self-assembly strategies. The ability to experimentally decompose DNA-CNT forces may open up a path to explore the fundamental interactions between nonpolar surfaces, solvent, and salts, especially given the diversity of the DNA-CNT system in terms of size (i.e., CNT diameter) and surface properties (i.e., CNT chirality and DNA sequence). Studies of different CNT chiralities and DNA sequences are also needed to correlate the measured interactions (especially the nonelectrostatic attraction) with their partitions in ATP systems and help elucidate the chemical and physical principles of DNA-CNT complexation and behaviors in ionic solutions.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpcc.6b04623.

Details on experimental measurements, discussion of the hydration force formalism, and comparisons and fits of force−distance curves (PDF)

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Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

We thank Erin Britt for help with experiments. The work was supported by the George Washington University.

DOI: 10.1021/acs.jpcc.6b04623
J. Phys. Chem. C 2016, 120, 13831−13835

ABBREVIATIONS

DNA, deoxyribonucleic acids; CNT, carbon nanotube; IEX, ion-exchange chromatography; ATP, aqueous two-phase; OSM, osmotic stress method; vdW, van der Waals

REFERENCES

(1) Wilner, O. I.; Willner, B.; Willner, I. DNA Nanotechnology. *Adv. Exp. Med. Biol.* **2012**, *733*, 97−114.

(2) Zhang, F.; Nangreave, J.; Liu, Y.; Yan, H. Structural DNA Nanotechnology: State of the Art and Future Perspective. *J. Am. Chem. Soc.* **2014**, *136*, 11198−11211.

(3) Baughman, R. H.; Zakhidov, A. A.; de Heer, W. A. Carbon Nanotubes - the Route toward Applications. *Science* **2002**, *297*, 787−792.

(4) Bergin, S. D.; et al. Towards Solutions of Single-Walled Carbon Nanotubes in Common Solvents. *Adv. Mater.* **2008**, *20*, 1876−1881.

(5) Hersam, M. C. Progress Towards Monodisperse Single-Walled Carbon Nanotubes. *Nat. Nanotechnol.* **2008**, *3*, 387−94.

(6) Tu, X.; Hight Walker, A. R.; Khripin, C. Y.; Zheng, M. Evolution of DNA Sequences toward Recognition of Metallic Armchair Carbon Nanotubes. *J. Am. Chem. Soc.* **2011**, *133*, 12998−3001.

(7) Tu, X.; Manohar, S.; Jagota, A.; Zheng, M. DNA Sequence Motifs for Structure-Specific Recognition and Separation of Carbon Nanotubes. *Nature* **2009**, *460*, 250−3.

(8) Zheng, M.; et al. Structure-Based Carbon Nanotube Sorting by Sequence-Dependent DNA Assembly. *Science* **2003**, *302*, 1545−8.

(9) Roxbury, D.; Tu, X.; Zheng, M.; Jagota, A. Recognition Ability of DNA for Carbon Nanotubes Correlates with Their Binding Affinity. *Langmuir* **2011**, *27*, 8282−93.

(10) Manohar, S.; Tang, T.; Jagota, A. Structure of Homopolymer DNA-Cnt Hybrids. *J. Phys. Chem. C* **2007**, *111*, 17835−17845.

(11) Ao, G. Y.; Khripin, C. Y.; Zheng, M. DNA-Controlled Partition of Carbon Nanotubes in Polymer Aqueous Two-Phase Systems. *J. Am. Chem. Soc.* **2014**, *136*, 10383−10392.

(12) Fagan, J. A.; Khripin, C. Y.; Batista, C. A. S.; Simpson, J. R.; Haroz, E. H.; Walker, A. R. H.; Zheng, M. Isolation of Specific Small-Diameter Single-Wall Carbon Nanotube Species Via Aqueous Two-Phase Extraction. *Adv. Mater.* **2014**, *26*, 2800−2804.

(13) Zhang, M.; Khripin, C. Y.; Fagan, J. A.; McPhie, P.; Ito, Y.; Zheng, M. Single-Step Total Fractionation of Single-Wall Carbon Nanotubes by Countercurrent Chromatography. *Anal. Chem.* **2014**, *86*, 3980−3984.

(14) Subbaiyan, N. K.; Cambre, S.; Parra-Vasquez, A. N. G.; Haroz, E. H.; Doorn, S. K.; Duque, J. G. Role of Surfactants and Salt in Aqueous Two-Phase Separation of Carbon Nanotubes toward Simple Chirality Isolation. *ACS Nano* **2014**, *8*, 1619−1628.

(15) Khripin, C. Y.; Fagan, J. A.; Zheng, M. Spontaneous Partition of Carbon Nanotubes in Polymer-Modified Aqueous Phases. *J. Am. Chem. Soc.* **2013**, *135*, 6822−6825.

(16) Qiu, X.; Khripin, C. Y.; Ke, F.; Howell, S. C.; Zheng, M. Electrostatically Driven Interactions between Hybrid DNA-Carbon Nanotubes. *Phys. Rev. Lett.* **2013**, *111*, 048301.

(17) Ke, F.; Qiu, X. Nanoscale Structure and Interaction of Condensed Phases of DNA−Carbon Nanotube Hybrids. *J. Phys. Chem. C* **2015**, *119*, 15763−15769.

(18) Leikin, S.; Parsegian, V. A.; Rau, D. C.; Rand, R. P. Hydration Forces. *Annu. Rev. Phys. Chem.* **1993**, *44*, 369−395.

(19) Stanley, C.; Rau, D. C. Evidence for Water Structuring Forces between Surfaces. *Curr. Opin. Colloid Interface Sci.* **2011**, *16*, 551−556.

(20) Parsegian, V. A.; Zemb, T. Hydration Forces: Observations, Explanations, Expectations, Questions. *Curr. Opin. Colloid Interface Sci.* **2011**, *16*, 618−624.

(21) Tardani, F.; Sennato, S. Phase Behavior of DNA-Stabilized Carbon Nanotubes Dispersions: Association with Oppositely-Charged Additives. *J. Phys. Chem. C* **2014**, *118*, 9268−9274.

(22) Tardani, F.; La Mesa, C.; Poulin, P.; Maugey, M. Phase Behavior of DNA-Based Dispersions Containing Carbon Nanotubes: Effects of Added Polymers and Ionic Strength on Excluded Volume. *J. Phys. Chem. C* **2012**, *116*, 9888−9894.

(23) Zheng, M.; Jagota, A.; Semke, E. D.; Diner, B. A.; McLean, R. S.; Lustig, S. R.; Richardson, R. E.; Tassi, N. G. DNA-Assisted Dispersion and Separation of Carbon Nanotubes. *Nat. Mater.* **2003**, *2*, 338−42.

(24) Parsegian, V. A.; Rand, R. P.; Rau, D. C. Macromolecules and Water: Probing with Osmotic Stress. *Methods Enzymol.* **1995**, *259*, 43−94.

(25) Rau, D. C.; Parsegian, V. A. Direct Measurement of the Intermolecular Forces between Counterion-Condensed DNA Double Helices - Evidence for Long-Range Attractive Hydration Forces. *Biophys. J.* **1992**, *61*, 246−259.

(26) Jin, H.; Jeng, E. S.; Heller, D. A.; Jena, P. V.; Kirmse, R.; Langowski, J.; Strano, M. S. Divalent Ion and Thermally Induced DNA Conformational Polymorphism on Single-Walled Carbon Nanotubes. *Macromolecules* **2007**, *40*, 6731−6739.

(27) Duggal, R.; Pasquali, M. Dynamics of Individual Single-Walled Carbon Nanotubes in Water by Real-Time Visualization. *Phys. Rev. Lett.* **2006**, *96*, 246104.

(28) Ghosh, S.; Dixit, H.; Chakrabarti, R. Ion Assisted Structural Collapse of a Single Stranded DNA: A Molecular Dynamics Approach. *Chem. Phys.* **2015**, *459*, 137−147.

(29) Kundagrami, A.; Muthukumar, M. Theory of Competitive Counterion Adsorption on Flexible Polyelectrolytes: Divalent Salts. *J. Chem. Phys.* **2008**, *128*, 244901.

(30) Grosberg, A. Y.; Nguyen, T. T.; Shklovskii, B. I. Colloquium: The Physics of Charge Inversion in Chemical and Biological Systems. *Rev. Mod. Phys.* **2002**, *74*, 329−345.

(31) French, R. H.; et al. Long Range Interactions in Nanoscale Science. *Rev. Mod. Phys.* **2010**, *82*, 1887−1944.

(32) Roxbury, D.; Mittal, J.; Jagota, A. Molecular-Basis of Single-Walled Carbon Nanotube Recognition by Single-Stranded DNA. *Nano Lett.* **2012**, *12*, 1464−1469.

(33) Ghosh, S.; Patel, N.; Chakrabarti, R. Probing the Salt Concentration Dependent Nucleobase Distribution in a Single-Stranded DNA−Single-Walled Carbon Nanotube Hybrid with Molecular Dynamics. *J. Phys. Chem. B* **2016**, *120*, 455−466.

DOI: 10.1021/acs.jpcc.6b04623
J. Phys. Chem. C 2016, 120, 13831−13835

Supplemental Information

Attractive Interactions between DNA-Carbon Nanotube Hybrids in Monovalent Salts

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1. Method of CNT dispersion

CNTs in powder form are sonicated in deionized water for 10 min with a Sonics VCX130 unit (3 mm-diameter probe, power of 5 w) before adding to DNA solutions. The mixture (DNA-CNT of ~1 mg/ml, 2:1 weight ratio of DNA/CNT) is further sonicated in ice bath for 4 hours at a power of 8 w, followed by centrifugation of 20,000×g for 3 hours to remove impurities and undispersed CNTs.

2. Force-distance curve measurements with the osmotic stress method (OSM)

The OSM measures force-distance relations by first setting the inter-molecular force/pressure at a specified value and then determining the inter-molecular distance. In this study, 80 µL DNA-CNT or dsDNA solution was added to 120 µL 15% w/w PEG8k (polyethylene glycol with molecular weight of 8000 Dalton) 300 mmol/L NaCl solution, and the mixture was vortexed and centrifuged at 20,000×g for 1 hour to get a pellet. This process left no detectable amount of DNA-CNT hybrids in the supernatant, and an image of the as-prepared pellet and the DNA-CNT dispersion is shown in Figure S1. We have shown in our previous study[1] that such pelleting process does not perturb the structure of DNA-CNT hybrids. Then the pellets were equilibrated in solutions of varied PEG8k concentrations (5 to 50% every 5% w/w) and NaCl concentrations (300, 500, 1000, 1500 and 2000 mmol/L) with one additional buffer exchange before x-ray diffraction (XRD) measurements. Note that an earlier version of the SG65i CNT product was used in our previous study,[1] and the force-distance curves from the two CNT batches showed the same

shape though with a 2-Å shift in inter-axial distances (Suppl. Figure S1). We attribute this to the likely difference in average CNT diameters.



Figure S1. Photo of a DNA-CNT pellet at the tube bottom after centrifugation (left tube) and the DNA-CNT dispersion (right tube).

3. Hydration force formulism with two exponential forces

The measured force by the osmotic stress method (OSM) is the thermodynamic potential of mean force between approaching surfaces and contains all contributions such as electrostatic, hydration, vdW, and entropic forces. As the OSM probes forces at close surface-surface separations down to a few Å, the approaching of the apposing surfaces also de-solvates both surfaces. The measured forces thus strongly depend on solvation energetics. For the case of dsDNA, electrostatic and solvation contributions dominate. It is however difficult to separate the electrostatic and solvation contributions to inter-dsDNA forces, as the charged molecular groups are also the initial causes for solvation. In order to elucidate the nature of inter-molecular forces at close separations, Parsegian, Rau, and coworkers have worked to examine forces between distinctly different surfaces (e.g., charged vs. non-charged, planar vs. cylindrical, and divergent surface chemistries of DNA, protein, lipid, or saccharides)[2-4]. All surfaces were observed to give a dominant exponential repulsive force at the closest separations (<~7 Å), as also illustrated by DNA-CNT and dsDNA forces in Figure 1 in the main text. Remarkably, this short-range repulsion can be

characterized by a universal exponential decay length of 2-3 Å independent of surface chemistry or salt concentration. At intermediate separations (e.g., 7 Å to a few nanometers), an exponential force of 4-6 Å decay length was observed between neutral surfaces or charged surfaces at high salts, and this force can be either attractive or repulsive depending on the surface hydration pattern. This led to the proposition of universal "hydration forces" that dominate at close molecular surface separations.

To explain the observed decay lengths of the exponential forces, a simplistic order parameter theory was developed by considering the energetics of water structuring with a correlation length λ ~4-6 Å[5-6]. In analogy to the decomposition of solute solvation energy into pure steric exclusion of solvent and solute-solvent interaction terms, the effects of two surfaces approaching each other can be separated into the de-solvation of each surface and the interaction between their respective hydration shells. The energetic cost of de-solvation gives rise to an "image-charge-like" repulsive force with a decay length of λ/2, and the interaction term has a decay length of λ and can be attractive or repulsive depending on whether the interaction enhances or weakens respective hydration structures. Analytically, the hydration force as from the order parameter theory can be written as, $f_d e^{-\frac{d}{\lambda/2}} + f_i e^{-\frac{d}{\lambda}}$, where d is inter-molecular separation and the force magnitudes f_d and f_i are determined by the hydration characteristics between the surfaces. For charged surfaces, there exists an additional electrostatic term that is always present but overwhelmed by hydration at short range. At longer range, the electrostatic contribution becomes significant, and it can be treated as the double-layer interaction potential, $f_e e^{-\frac{d}{\lambda_D}}$, where λ_D is the Debye length. Fitting experimental curves with three exponentials above results in excellent agreements but thus obtained force parameters have strong correlations, especially between the f_i and f_e, as λ and λ_D are rather similar at medium to high salts. Due in part to the limited range of experimental data, an effective interaction, $f_{eff} e^{-\frac{d}{\lambda_{eff}}}$, is often empirically used to characterize the long range force. The decay length λ_{eff} closely follows the trend of the Debye length λ_D at low salts and decreases with increasing salts. At high salts, λ_{eff} approaches the water correlation length λ of ~4-6 Å. Hence, experimental inter-molecular forces can be parameterized by two exponentials as $F(d) = f_d e^{-\frac{d}{\lambda/2}} + f_{eff} e^{-\frac{d}{\lambda_{eff}}}$ with f_d, f_i, and λ_{eff} as the unknowns determined by curve fitting.

4. Additional figures

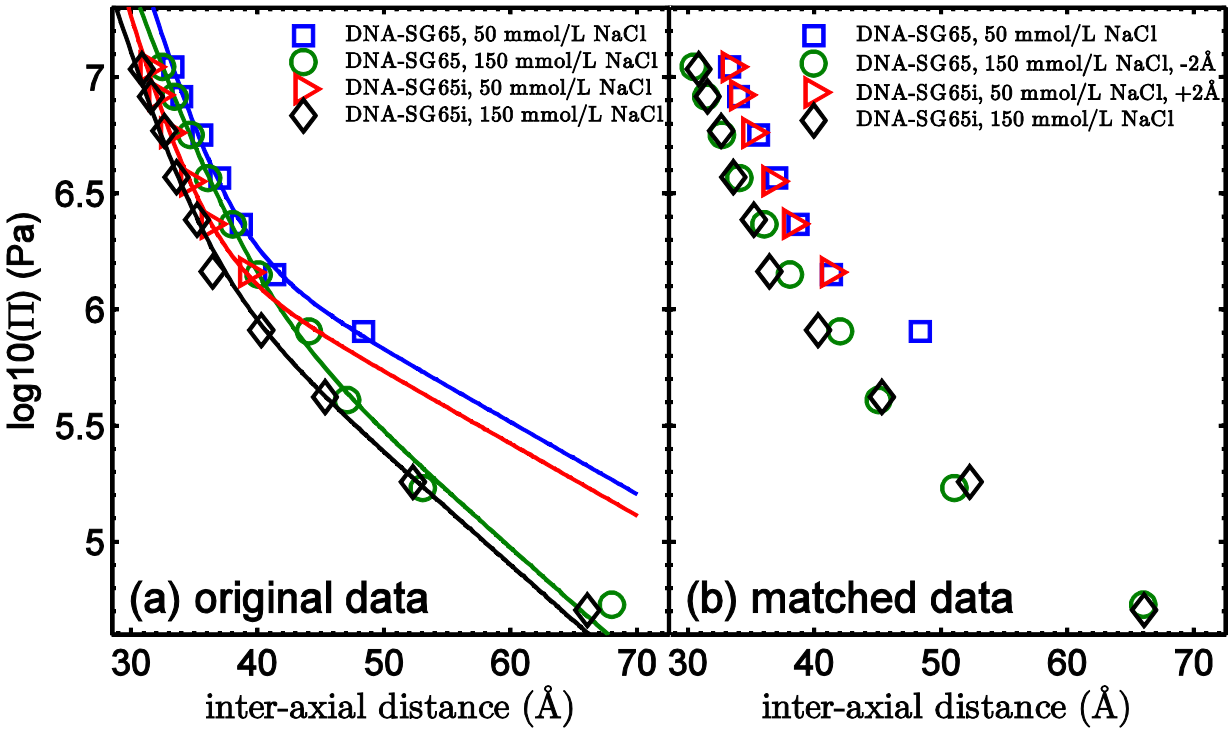


Figure S2. Comparison of the force-distance curves at low salts from two different batches of CNTs. The same $(GT)_{20}$ DNA was used to disperse both CNT types. One batch of CNT is the CoMoCAT SG65 from Southwest NanoTechnologies (DNA-SG65) we previously studied;[1] The other is the CoMoCAT SG65i from Southwest NanoTechnologies (DNA-SG65i) used in this study. (a) The original force-distance curves are shown, and the lines shown in the same color are the corresponding fits using two exponentials. We observe that DNA-SG65 and DNA-SG65i hybrids show different force-distance curves. However, the major difference between the two DNA-CNT hybrids appears to be a shift in the inter-axial distance, which we attribute to the different average diameters of the two CNT types. (b) A shift of 2 Å in the inter-axial distance is found to match the DNA-SG65 and DNA-SG65i force curves. Note that, for clarity purposes, the DNA-SG65 curve at 150 mmol/L NaCl is shifted by -2 Å and the DNA-SG65i curve at 50 mmol/L NaCl is shifted by +2 Å.

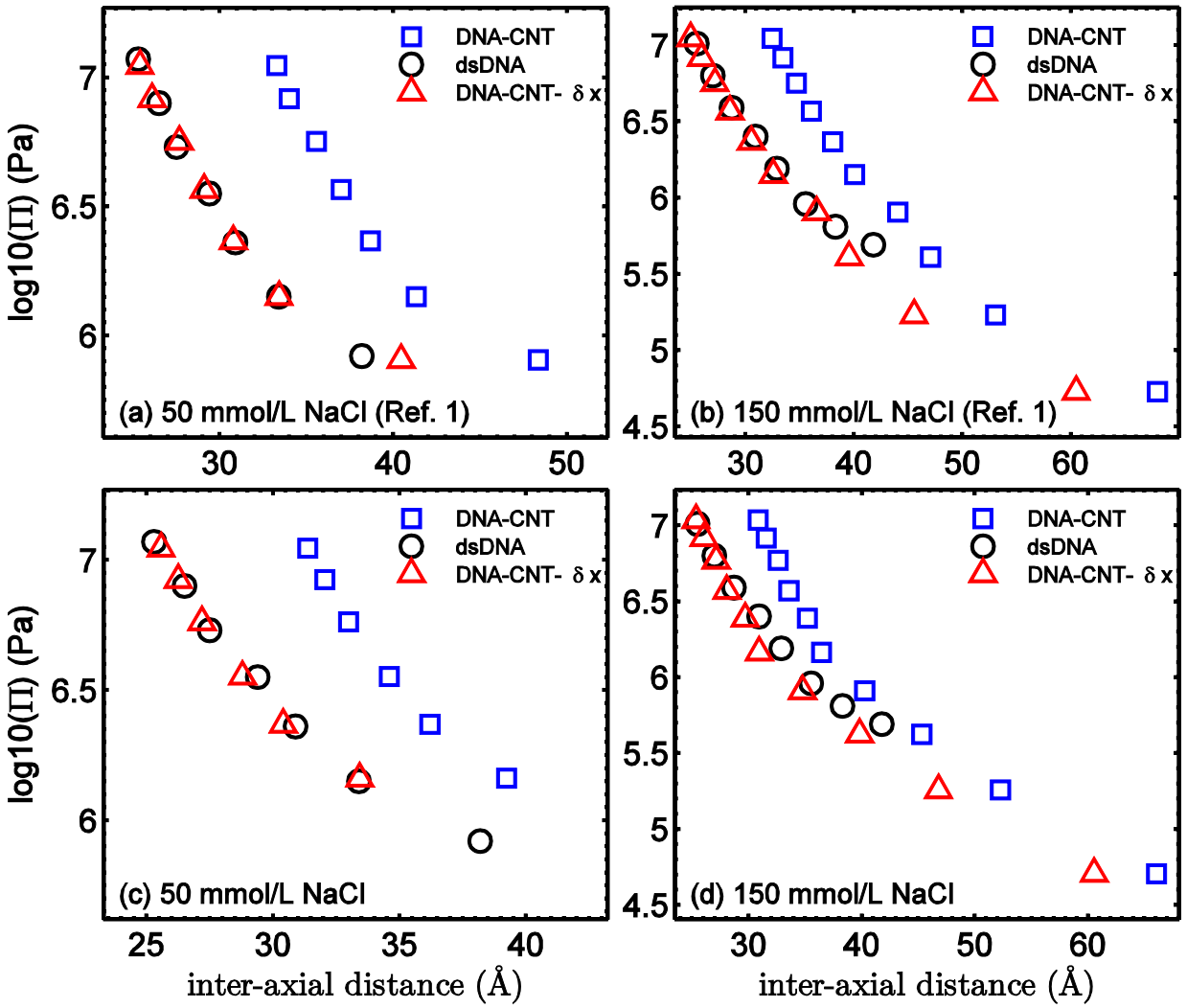


Figure S3. Comparisons of the force curves of DNA-CNT hybrids and dsDNA at low salts of 50 and 150 mmol/L NaCl. In each panel, the original DNA-CNT force curve is shown as open square; the original dsDNA force curve is shown as open circle; and the original DNA-CNT curves is then shifted by a constant offset in the inter-axial distance and shown as open triangle. Panels (a) and (b) show the 50 and 150 mmol/L NaCl data from Ref. 1, respectively where the CoMoCAT SG65 CNT was used. Panels (c) and (d) show the 50 and 150 mmol/L NaCl data from the CoMoCAT SG65i CNT used in this study. All comparisons show essentially the same trends from DNA-CNT hybrids and dsDNA at low salts.

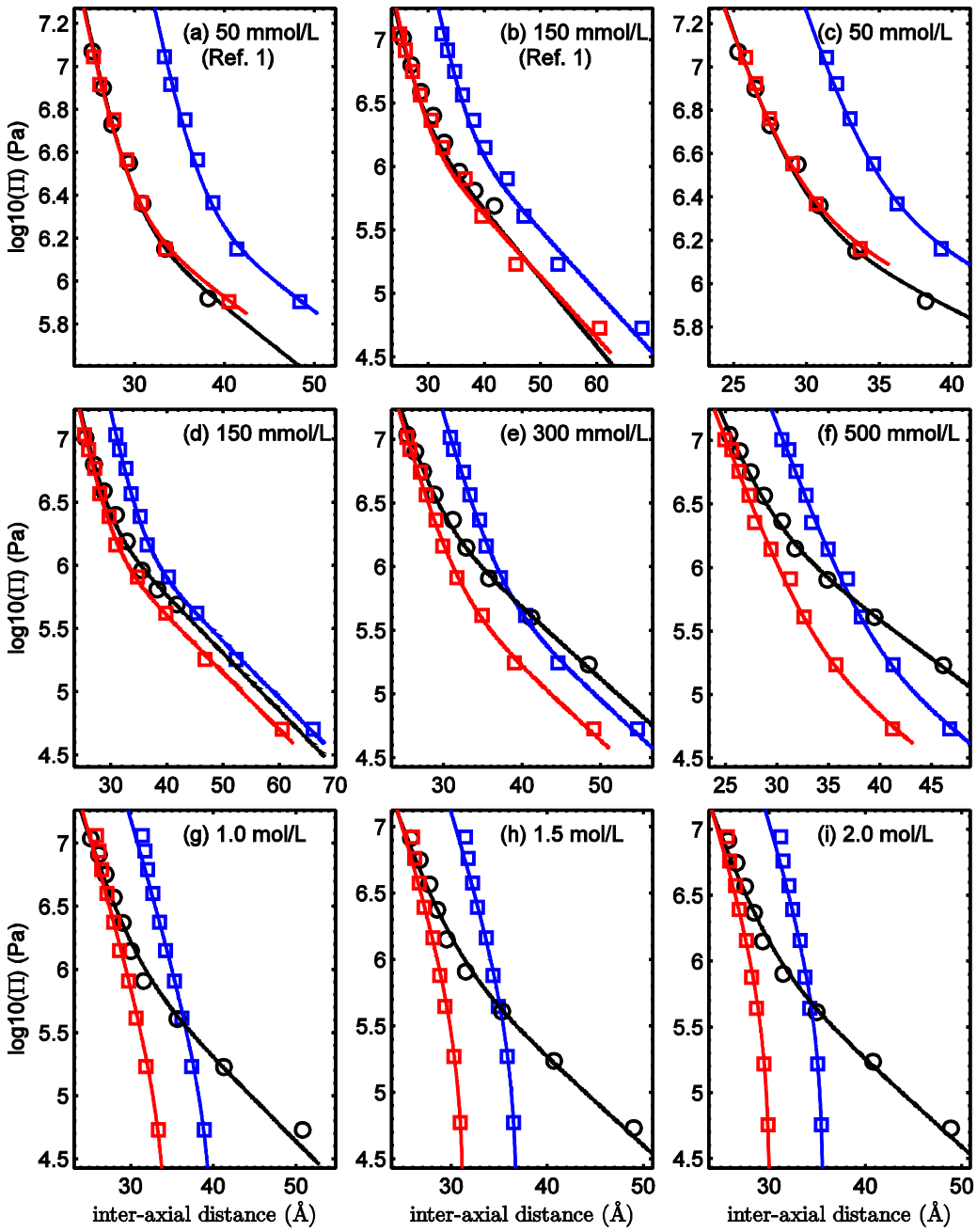


Figure S4. Fits to DNA-CNT and dsDNA force-distance curves based on the double-exponential forces and illustrations of the matches between DNA-CNT and dsDNA force curves by offsetting their difference of 5.5 Å in diameter. In each panel, the original DNA-CNT experimental data (open square) and its fit (solid line) are shown in blue, while the dsDNA experimental data (open circle) and its fit (solid line) are shown in black . The offsetted DNA-CNT data and fit are shown in red. Note that panels (a) and (b) show the data previously published in Ref. 1 by offsetting their difference of 7.5 Å in diameter (see more discussions in Figure S1).

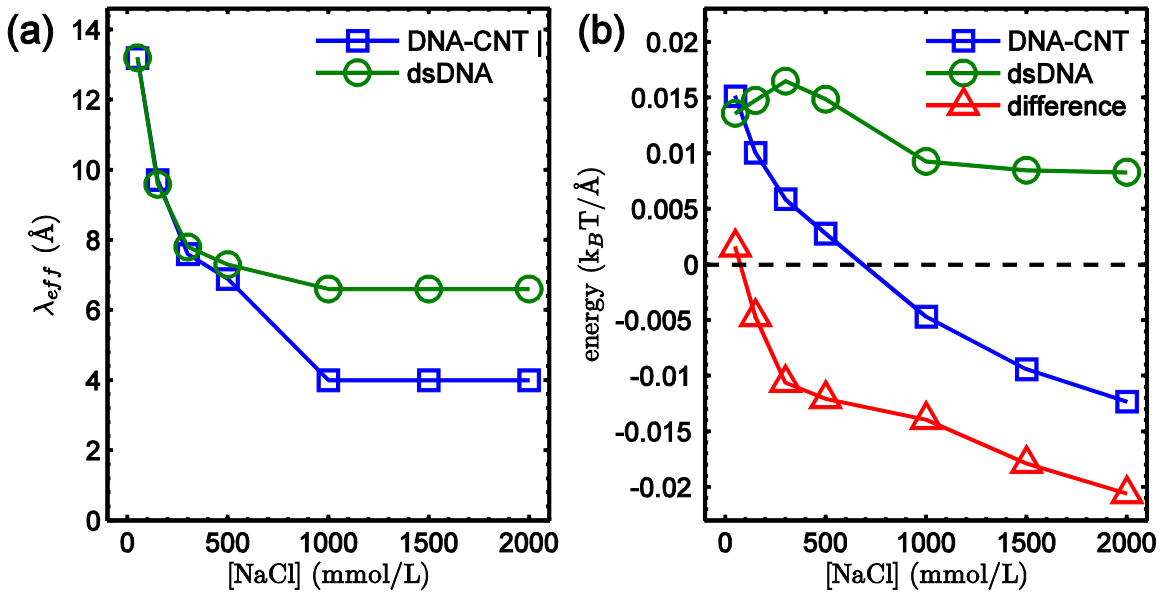


Figure S5: Parameters and energetics of the second exponential force from the fits to DNA-CNT and dsDNA forces. (a) The decay length λ_{eff} of the second exponential force, $f_{eff}e^{-\frac{d}{\lambda_{eff}}}$. At [NaCl] < 100 mmol/L, the obtained effective decay lengths from DNA-CNT and dsDNA forces are similar, while the force magnitudes vary significantly (see Figure 2d in the main text). At [NaCl] ≥ 1000 mmol/L, the λ_{eff} values for dsDNA reach a plateau as the forces become largely salt independent. For DNA-CNT forces at [NaCl] ≥ 1000 mmol/L, their "convex" shapes (see Figure S3) suggest that the second exponential force is attractive and is dominated by the hydration attraction. We thus set the decay length as the water correlation length of 4 Å within the framework of hydration force formulism. (b) Interaction energies due to the second exponential force alone for DNA-CNT hybrids and dsDNA. The interaction energy is obtained by integrating the force curves between 5 and 10 Å surface separations that are directly probed by experiments. As expected, the trends of energetics follow that of the force/pressure magnitude data shown in Figure 2d in the main text.

REFERENCES

1. Qiu, X.; Khripin, C. Y.; Ke, F.; Howell, S. C.; Zheng, M., Electrostatically Driven Interactions between Hybrid DNA-Carbon Nanotubes. *Phys. Rev. Lett.* **2013**, *111*, 048301.
2. Stanley, C.; Rau, D. C., Evidence for Water Structuring Forces between Surfaces. *Current Opinion in Colloid & Interface Science* **2011**, *16*, 551-556.
3. Parsegian, V. A.; Zemb, T., Hydration Forces: Observations, Explanations, Expectations, Questions. *Current Opinion in Colloid & Interface Science* **2011**, *16*, 618-624.
4. Leikin, S.; Parsegian, V. A.; Rau, D. C.; Rand, R. P., Hydration Forces. *Annu. Rev. Phys. Chem.* **1993**, *44*, 369-395.

5. Gruen, D. W. R.; Marcelja, S., Spatially Varying Polarization in Water. A Model for the Electric Double Layer and the Hydration Force. *Journal of the Chemical Society, Faraday Transactions 2: Molecular and Chemical Physics* **1983**, *79*, 225-242.
6. Marčelja, S.; Radić, N., Repulsion of Interfaces Due to Boundary Water. *Chem. Phys. Lett.* **1976**, *42*, 129-130.